

Final Year Project



Department of Information Technology

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SMART AI LANTERN



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*Submitted for the partial fulfilment of BS Information Technology degree to the
Faculty of Sciences*

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ABSTRACT

For that, the specific goal of the design of the Smart AI Lantern project is: “to create a smart lamp that has AI for features such as voice recognition, ambient color and adaptive brightness using Body Sensors”. Most smart lighting systems currently face issues in the integration of AI in the systems as well as the user interfaces. To solve these challenges, we developed the new convenient and easy to use lantern that can be controlled by voice and has the feature of automatic brightness control if the person is inside the lantern or not.

The starting point of this research is the analysis of the present smart home technologies and the use of AI in human-computer interface interactions. Subsequently, we came up with and manufactured the Smart AI Lantern which has, integrated sophisticated voice recognition application and body sensors. We continued testing the lantern during development while incorporating feedback from users so that it matched their needs. The main areas of concern for us were prompt response to voice commands and accuracy of body sensors with minimal delays.

As far as reliability is concerned, throughout the testing and validation stages, our team exposed Smart AI Lanterns to various real life situations. We have successfully engineered a system that promptly responds to voice commands while adjusting itself automatically depending on whether such conditions.

CERTIFICATE

Dated _____

Final Approval

It is certified that project report titled “**Smart AI Lantern**” submitted by **Aena Azhar**, for the partial fulfillment of the requirement of “**Bachelor’s Degree in Information Technology**” is approved.

COMMITTEE

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ACKNOWLEDGMENT

In the name of Allah, the most gracious and most merciful. We are grateful to Him who blessed us the capability to do this project.

“It is not possible to prepare project report without assistance & encouragement of our respective supervisor Ma’am Sidra Khudadad. This one is certainly no exception.”

Or

“Success isn’t a matter of being the best & winning the race. Success is a Matter of handling the worst & still finishing the race.”

First of all, thanks to our parents for giving encouragement, enthusiasm and invaluable assistance to us. Without all of this, we might not be able to complete this project.

Secondly, we are extremely thankful to our supervisor Ma’am Sidra Khudadad for her valuable guidance and support on completion of this project. Their valuable suggestions, guidance and support on completion of this project was just beyond measure. Her proficiency, dedication, honesty and continuous encouragement helped us a lot.

We are also thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during the project work. Our special gratitude also goes to whole IT faculty for their valuable guidance, fruitful discussions.

As truly said: “DONE, is better than perfect.”

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CHAPTER 1

INTRODUCTION

(Introduction to Project and Project Report)

1.1 Introduction:

1.1 Introduction:

With the recent developments in technology, Artificial Intelligence (AI) has been integrated into numerous everyday items to make them more innovative. In this light, we showcase a high-tech project that seeks to develop an intelligent lamp which merges voice recognition, ambient color adjustment and adaptive brightness control via body sensors. This is a smart lamp which epitomizes the fusion of AI and user-centered design implying that it offers users a unique and intuitive interaction model so as to easily manage their illumination environment. It can turn on/off based on voice commands.

1.2 Problem Statement:

Despite the abundance of smart appliances, there still exists no clear-cut solution for automatic home lighting systems. Most of current technologies do not integrate seamlessly with one another and also lack user-friendly interfaces making it difficult for users to take full advantage of AIs potential on an everyday basis.

Our project takes care of these limitations by designing a smart lamp capable not only respond to voice commands but also having body sensors used in controlling its brightness adaptively thus enabling it to provide holistic and user-centric lighting experience. To sum up, our research identifies these problems and aims at offering a unified and intelligent approach that makes interactions between people and their lighting devices simpler.

Vision

Visualize a future where lighting settings effortlessly adjust to our requirements and desires, resulting in an enhanced level of comfort, security, and productivity. Public areas, offices, and homes would all feel more responsive, promoting productivity and well-being.

Issue Statement

The main problem is that customers are unable to properly employ AI-driven lighting solutions because current smart lighting systems are difficult to use and integrate. As a result, smart home environments lack cohesiveness and provide a fragmented user experience. Improving user happiness and the widespread deployment of smart lighting systems need addressing this issue.

Method

Our project aims to solve this problem by creating a voice-activated smart lamp with body sensor adaptability for adjustable brightness management. This is the procedure we'll use:

1. What issue or requirement exists?

The issue is the absence of a single, user-friendly smart lighting system that makes full use of AI to enable smooth integration and simple operation.

2. Who is in need or has the problem?

This requirement applies to homeowners and anyone who use or wish to utilize smart home technologies on a daily basis. This includes anyone trying to control their house lighting in a more practical and effective manner.

3. Why is it important to solve?

The reason this problem needs to be solved is that, by making users' homes more pleasant and manageable, it can significantly enhance their quality of life on a daily basis. It also improves user experience overall and contributes to energy savings.

4. If the issue is not resolved, what will happen?

Users' frustration and discomfort with their smart lighting systems will persist if the issue is not resolved. The potential advantages of AI, including lower energy costs and, will be lost on them.

1.3 Aim & Objectives

Aims:

This project's objectives are to:

1. Create a smart lamp that improves user experience by combining body sensors and voice recognition for adaptive brightness management.
2. To design a logical and well-organized house lighting system.

Objectives:

1. To investigate and evaluate current AI applications in house lighting systems and smart home technologies.

Review the literature to find out where smart lighting solutions are at the moment and to pinpoint any integration and user experience gaps.

2. To create a voice-activated smart lamp using cutting-edge technology.

Create and put into use voice recognition algorithms that let users turn on and off the lamp and alter its color with simple voice commands.

3. To incorporate body sensors for control of adaptive brightness.

Develop and implement a system using body sensors which can sense the presence of user and change the lamp brightness in order to provide the best lighting throughout.

4. To develop and improve the user interface such that it is able to communicate without any problem.

Make a user-friendly interface that is intuitive enough to allow interaction with customers.

5. To collect user input and test the smart bulb in real-world situations.

Conduct tests on how the lamp functions, how easy it is for users to use it, and how responsive users are. Gather information for making necessary modifications or adding more features.

6. To assess the smart lamp's overall impact and energy efficiency.

Look at what power is used by this lamp in contrast to whether adaptive brightness management could save you money. In order to show its benefits compare these results with those derived from conventional lighting systems.

1.4 Proposed System

The Smart AI Lantern – this is a proposed system – is a state-of-the-art intelligent lamp that brings together advanced technology for better comfort while changing home

illumination at once. This intelligent lamp overcomes problems faced by current smart lighting systems through integration of body sensors and voice recognition to adapt brightness across rooms.

Voice Recognition:

Users can operate Smart AI Lantern by simply speaking commands to it. This feature improves user convenience.

How the Issues Are Solved:

Integrated Effortlessly:

The Smart AI Lantern is a comprehensive and easy to use home lighting system that integrates speech recognition and body sensors. This integration provides a single interface to the user since the current smart home solutions are often fragmented and complicated.

Better User Experience:

The lamp also helps in the usage of lighting fixtures since users can control the amount of light to their preference. This makes the use of AI driven smart home products more popular and enhances the satisfaction of the customers.

Energy Efficiency:

Lighting control is optimized to ensure that energy is used efficiently through brightness control that adapts to the surrounding environment and occupancy. As a result of this, it not only reduces the amount of electricity consumed but also encourages environmentally friendly practices in the society.

Designing with the user in mind:

The Smart AI Lantern alters the way people engage with their homes through the use of design concepts such as automatic control and easy-to-use interfaces. By incorporating the advanced AI technology, it promotes a more comfortable and efficient living space.

1. 5 Development Methodology

The development methodology of Smart AI Lantern project includes a systematic approach that is designed to address the mentioned challenges and achieve the objectives of the project.

The methods used consist of:

1. Analysis and Research:

Carrying out a market research of the AI applications in the lighting systems and the existing smart home technologies in order to reveal the market advantages and drawbacks.

Conducting a literature analysis to analyze consumers' preferences, new technologies, and trends in the smart lighting design.

2. Prototyping and Design:

Specify firm design details based on the user needs and the outcomes of the analysis. Building Smart AI Lantern models that would allow for proper testing of various parts of the model inclusive of the body sensor, ability to recognize speech, and the ability to adapt to brightness levels.

3.Integration of Technology:

For instance feasible and state of the art technologies like the body sensors, voice recognition software's, microcontroller systems to allow the light to have inherent ease of operation and connectivity. Making sure that mechanisms of various types of facilities in the computer, or software programs function well and cohesively to optimize required performance and functionality for the end user.

4.User Interface Development:

Designing a user-friendly interface that will make the settings change of the lighting and its control rehearsed. Enhancements of the user interface and feedback data and actual data about the usability of the system through design-reviews and usability tests.

5.Validation and Testing:

Conducting the numerous testing stage so as to determine the efficiency of the Smart AI Lantern under variety situations or environments.

Executing the functional testing to make sure that sensor detections, voice commands, and adaptive lightning alterations are accurate, fast, and reliable.

6.Development Iteratively and Optimization:

Applying the repetition development process to address issues, enhance features, and boost the system's efficiency regularly.

Ongoing modifications that integrate evaluation results and users' feedback to optimize the smart lamp regarding its use, performance and reliability.

7.Reporting and Documentation:

Documents, having recordings of all steps from the development process: technical characteristics, decisions made, tests performed, and results obtained.

Preparation of a well documented report that explains the strategy, challenges encountered, measures adopted and achievements made throughout the project.

1.6 Resource Requirement

For the proper functioning of the Smart AI Lantern project and its additional features, it

is necessary to use both software and hardware components.

1.7 Social Benefits

The following significant advantages of the Smart AI Lantern project add to overall social benefits as well as the delight of individual users:

The project's benefits are as follows:

1. Enhanced User Experience.

Voice recognition and adaptive brightness control facilitate a direct interaction between the users and home lighting to create spaces that are much more comfortable and convenient.

2. Energy efficiency:

Energy is saved due to adaptive brightness control feature which automatically reduces the brightness depending on the surrounding conditions and the presence of a person. This helps to reduce the tariff of electricity and promote the use of power that is friendly to the environment.

3. Equality and Accessibility:

With the help of the voice control options, a specific category of people with disabilities or limited mobility can manage the light in their homes using the Smart AI Lantern much easier.

4. Technological Advancement:

The project fosters further development of AI combined with user-oriented ideas and concepts, and emphasizes on smart home. Sustainable intelligent living environments are being developed because of this.

5. Environmental impact:

It is environmentally sustainable since it eradicates carbon footprints specific to home energy usage and promotes energy-saving measures with automatic dimming of lights.

Project Recipients:

Homeowners and Residents: The Simple and intuitive design of the Smart AI Lantern is advantageous to users who want more ease and self-organization of houses and rooms illumination.

People with impairments: Voice control of the functions and adjustable sensors help in fulfilling the requirements of the physically impaired persons in order to gain more functional mobility within their homes.

Technology Enthusiasts and Developers: The creative approach and execution of the project provide insights and inspiration for professionals and enthusiasts in the domains of artificial intelligence, Internet of Things, and smart home technologies.

Society at Large: By lowering overall energy consumption and promoting sustainable living habits, the adoption of energy-efficient smart home solutions leads to wider societal advantages.

1.8 Report Layout

1. Introduction

Technology has brought many smart items into our lives, like our project: a smart and/or an illumination device. This lamp compares with voice commands, changes the colors of the room, and changes the brightness with the body sensors. Its goal is to assure the user that handling home lighting with the help of AI is easy.

2.Problem Statement

Most of the smart home systems available in the market have numerous interfaces and are interlinked, but the systems are not very efficient. Our lamp does this by integrating voice controls and body motion for easy control of light. Thus, it is used to decentralize home lighting.

3.Vision

Think about lights that turn on if required; this will make homes and offices more comfortable, and the employees more productive.

Issue Statement

Smart lighting systems of the present day are complicated and incompatible greatly diminishing the user experience. I believe that fixing this will result in a boost of user satisfaction as well as the utilization of smart lighting.

Method

They will include a voice-activated lamp where the body sensors will be used to control brightness of the lamp. It will be conducted a study on the current solutions, development of the lamp, prototyping, as well as a collection of feedback for improvement.

Aims & Objectives

Aims:

- 1.Introduce a smart lamp that can be controlled through voice and the sensor on the human body.
- 2.Create the home lighting system.

Objectives

1. Investigate the present AI lightning solutions.
2. Introduce voice activation for the lamp.
3. Incorporate active dimming sensors into your body.
4. Create an easy-to-use interface.
5. Apply the lamp with real users.
6. Evaluate the energy conservation and the degree of influence.

Proposed System

Smart AI Lantern

Used voice control and body sensors, to control brightness of the light in an easy way that does not consume much power.

Solves Issues:

- i. In this regard, the benefits of using social media include: They are easy to integrate and to use.
- ii. Enhancing user experience.
- iii. Promoting energy efficiency.
- iv. Therefore, the main principle of such approach is to concentrate on usability of the product.

Social Benefits

Benefits Include:

- i. Greater satisfaction for the user through simple lighting control.
- ii. Reduced energy consumption due to adaptive brightness of the displays.
- iii. Accessible routes for clients in wheel chairs and or with other forms of disability.
- iv. Modernization in AI and smart home integration.
- v. Environmental positivism achieved through the conservation of energy.

Project Recipients:

Beneficiaries of this project include home owners, persons with disability, technologically inclined individuals as well as the public society at large.

CHAPTER 2
BACKGROUND & EXISTING WORK
(Literature Review, Existing Work)

2.1 Background Study

Artificial Intelligence (AI) breakthroughs have been a chief riding force behind the speedy growth of clever home automation, specially in terms of enhancing human interactions and studies with technology. One extremely good component of this improvement has been the shift from voice-activated gadgets to human-machine interfaces which can be intuitive. In spite of those traits, there's still a dearth of studies into the combination of state-of-the-art capabilities like body sensor-based adaptive brightness control and dynamic shade adjustment.

- **Message Ritual (Main paper)** This LED lamp notifies messaging on another Twitter or Facebook, but it integrates more crucially. This light eavesdrops on the conversations in a house, and when it hears something interesting, turns exchange into generated poems. And the housemates are written poems to in daily text messages. The researchers looked at how people interacted with this lamp over a period of two weeks and what the experience was like during that time in their homes. The paper reflects on what they learned and discusses how this new view can inform future research in Human-Computer Interaction.

Editors Albrecht Schmidt, Kaisa Väänänen, + **Association for Computing Machinery** New York NY United States **Conference CHI '23: CHI Conference on Human Factors in Computing Systems** April 23 — →

Editors: Albrecht Schmidt, Kaisa Väänänen, + **6** **Publisher:** Association for Computing Machinery New York NY United States **Conference: CHI '23 (CHI)**

Date April 23 - April 28, **ISBN** 978-1-4503-9421-5 **Series** Pages Preview **Citation String** Ref Ammann et al.

- This paper discusses our relationship with technology and how we are bonding more closely to the gadget is given in this, like a lamp. (sources: This Lamp Hears You! This lamp hears what people talk about at home, then they distill it and make poems out of this. Every day it sends these little poems to people in occasional good morning text messages. For two weeks, the researchers examined how people felt about living with this lamp. The paper mulls over what they found and how this perspective on innovation may be

AI-Based Dimming Street Light with Energy Efficiency Url: Pathak, A., Bala Murugan, M.S., Rajagopal, M.K. (2023). **Smart Dimming Streetlight with Energy Optimization** AI In: Subhashini N, Ezra MAG, Liaw SK (eds) Sangam Digital Media Pvt. Ltd., Bauer J.M.C.-P.L.T.H.R.V.K.A.E.S.F.D.Y.Q.C.comp) **Advanced Information and Knowledge Processing**, vol 65. **Lecture Notes in Electrical Engineering**, vol 966 Springer, Singapore.

- Message Ritual - A Smart Lamp that Eavesdrops on Home Conversations And Turns Them into Poems and Good Morning Texts People used it after that for a couple of weeks and this paper is about what happened, why you should care (as other potential authors), and to whom.

Aditya Kolakoti, Ruthvik Chandramouli Department of Mechanical Engineering, Raghu Engineering College(A), Visakhapatnam Submission 25-03-2023; revision 27-04-2023; Acceptance 03-05-2023; Online publication 12-05-2023; Final published version 12-05-2023.

- In fact, the proposed study helps in presenting a method that can be easily used by people particularly those with color blindness to look at the colors. They utilize a small thing you call ESP32; the contraption is linked to the Internet to convey color information. The picture, redrawn in simpler terms, is trying to convey the information as follows The device All encodes the colors and has names for them Color and also the device has a Speech Synthesizer and the device talks about the color. Different from the other methods, this method informs color details explicitly and this is important for people who have color vision problems. Studies revealed that it is useful in many situations and correctly identifies a vast amount of shades. It is also connected to the Internet which users can frequently check the colour information in the case that it has been modified.

Voice Guidance System for Color Recognition Based on IoT Wen-Tsai Sung, Guan-Rong Chen¹ and Sung-Jung Hsiao^{2,1} **Department of Electrical Engineering, National Chin-Yi University of Technology, Taichung, 411030, Taiwan** ² **Department of Information Technology, Takming University of Science and**

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he or she is more specifically an Anthropologist. Email: sungjung@gs. takming. edu.
tw Received: Recieved: 10 February 2022; Revised: 25 March 2022; Accepted: 13
May 2022.

2.2 Literature overview

2.2.1 Integration of AI in smart domestic era AI's role in clever domestic gadgets has been pivotal, enhancing functionalities past simple automation studies have confirmed AI's functionality to investigate user behavior, are expecting choices, and autonomously adjust settings. This functionality no longer most effective complements comfort and efficiency but also lays the inspiration for personalized and adaptive consumer reports.

2.2.2 Consumer-Centric layout concepts effective clever domestic systems prioritize consumer enjoy via intuitive interfaces and customizable controls. Literature emphasizes the significance of seamlessly integrating generation into users' each day lives, providing flexibility and simplicity of use. user-centric layout standards make sure that smart home answers aren't handiest practical but also enhance typical comfort and pride.

2.2. The three energy efficiency issues Adaptive brightness control, which employs both the natural lighting and occupancy sensors, appears as a viable way of reducing the energy consumption while maintaining the users' satisfaction. They are sustainable techniques and they help in creating strong forms of aid use within the residential settings. Most attempts to manage the intake of power within smart homes have been directed towards the control of wise lighting fixtures.

Venture Justification and goals:

The outstanding innovation of clever AI Lantern undertaking is to fill the gaps seen in the present day smart domestic lighting systems via the use of AI.

This venture targets to:

Enhance comfort: Introduce voice manipulate that can be easily operated.

Improve strength efficiency: adaptive brightness? try to implement adaptive brightness manage to reduce the power consumption.

Beautify consumer experience: provide vivid spectral change and flexible lighting perfect fit to consumer options and presence.

Functional requirements:

1. **Voice manipulate:** permit voice popularity for controlling on/off capability and adjusting coloration settings.
2. **Adaptive Brightness manage:** combine frame sensors to locate occupancy and regulate brightness levels dynamically.
3. **Color Adjustment:** provide alternatives for dynamic coloration adjustments primarily based on consumer preferences and environmental conditions.
4. **Real-Time Responsiveness:** make sure brief response instances to voice instructions and sensor inputs for a continuing consumer enjoy.
5. **Compatibility:** make certain compatibility with existing clever domestic ecosystems for clean integration.

Non-functional requirments:

1. **person Interface:** layout an intuitive and consumer-friendly interface reachable to users of various technical expertise.
2. **Reliability:** make sure high system reliability with minimum downtime or disruptions in capability.
3. **Protection:** put in force robust security measures to guard consumer facts and save you unauthorized get right of entry to.
4. **Scalability:** design the device to scale correctly with ability future enhancements or extended person needs.

CHAPTER 3

SYSTEM MODEL / PROPOSED SYSTEM

3.1 Flow Chart Diagram

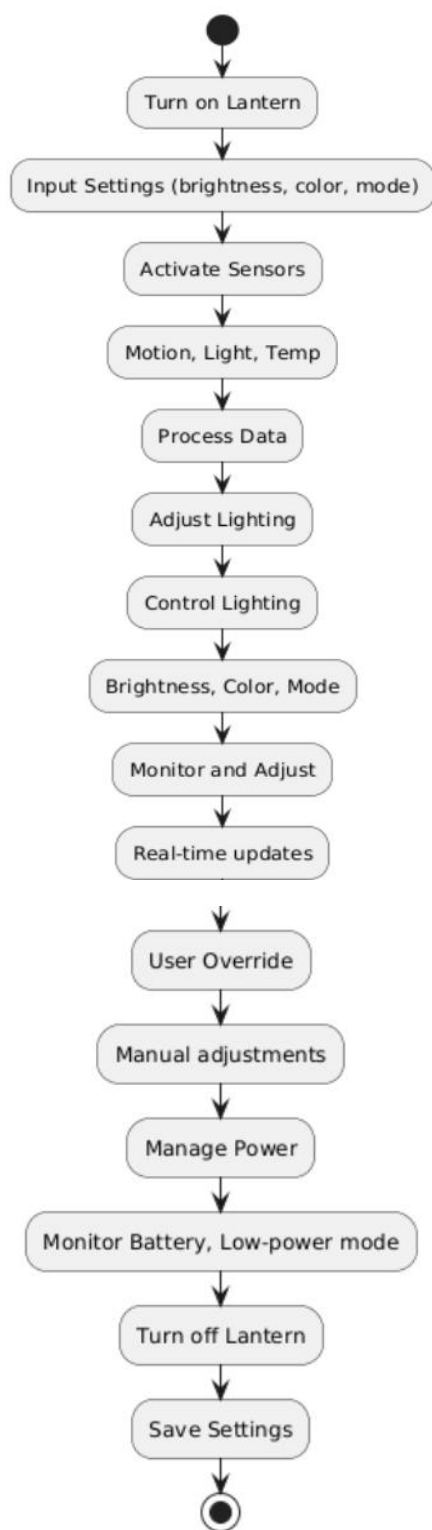


Figure 3.1 Flow Chart Diagram

3.2 Use Case Diagram

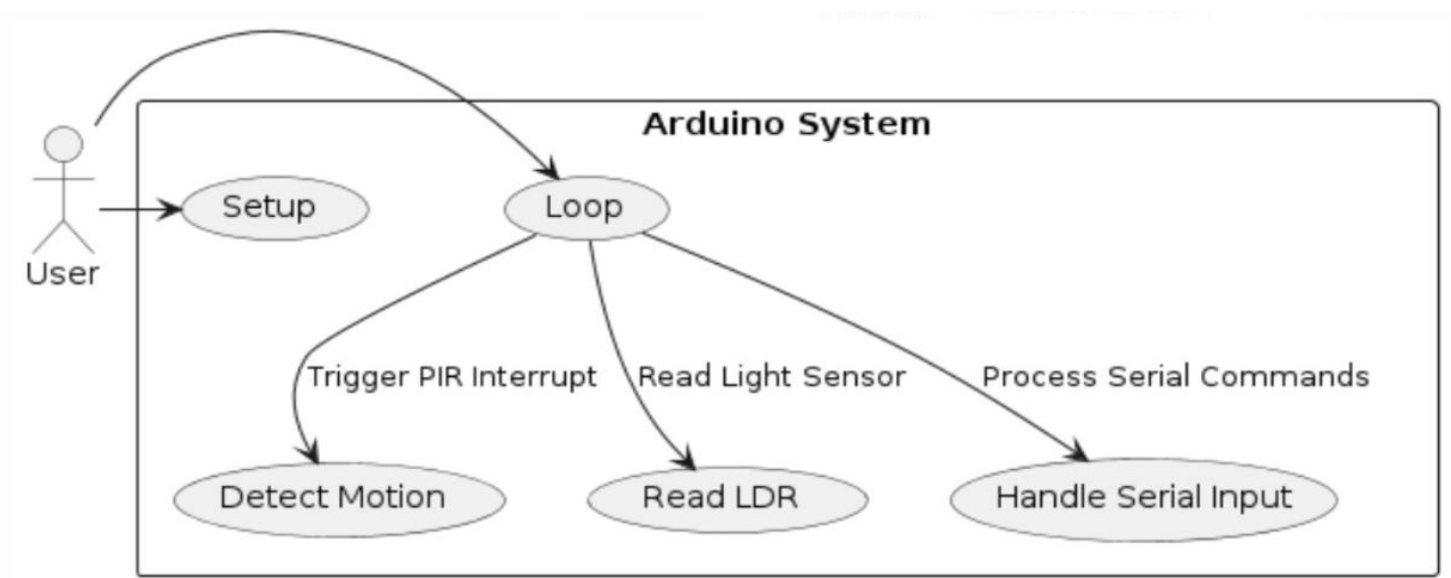


Figure 3.2 Use Case Diagram

3.3 Class diagram

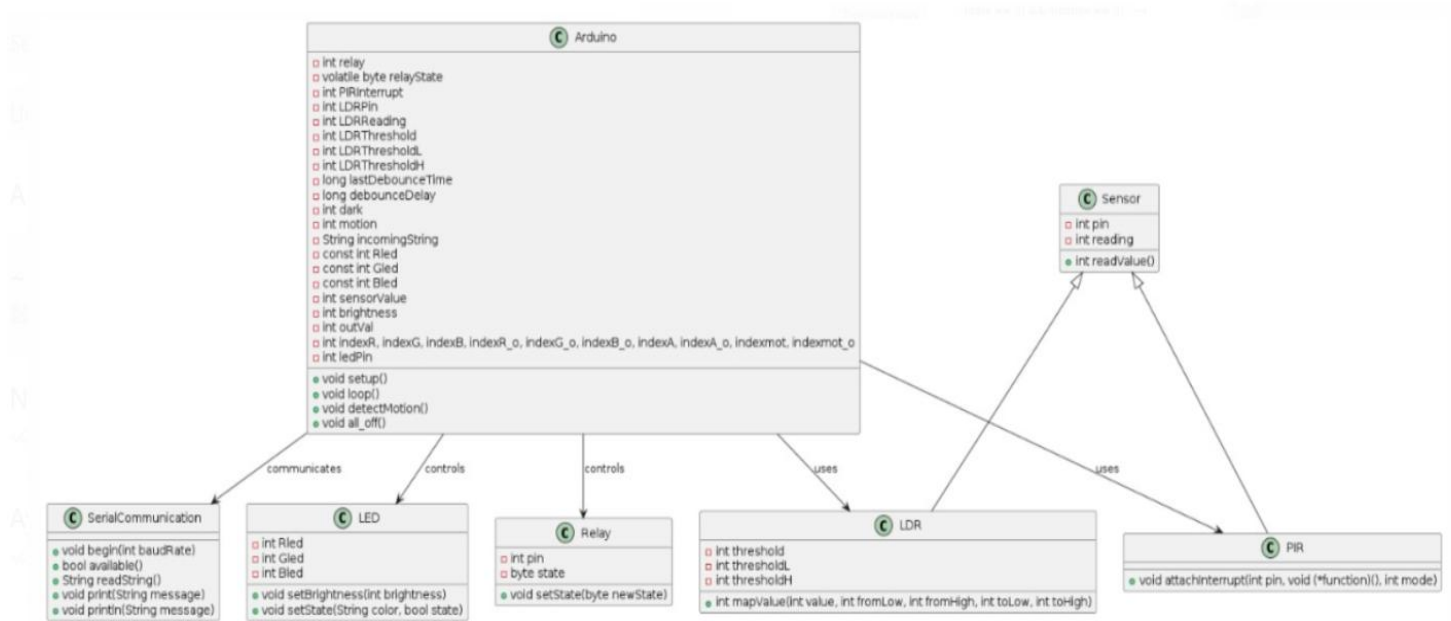


Figure 3.3 Class Diagram

3.4 ERD Diagram

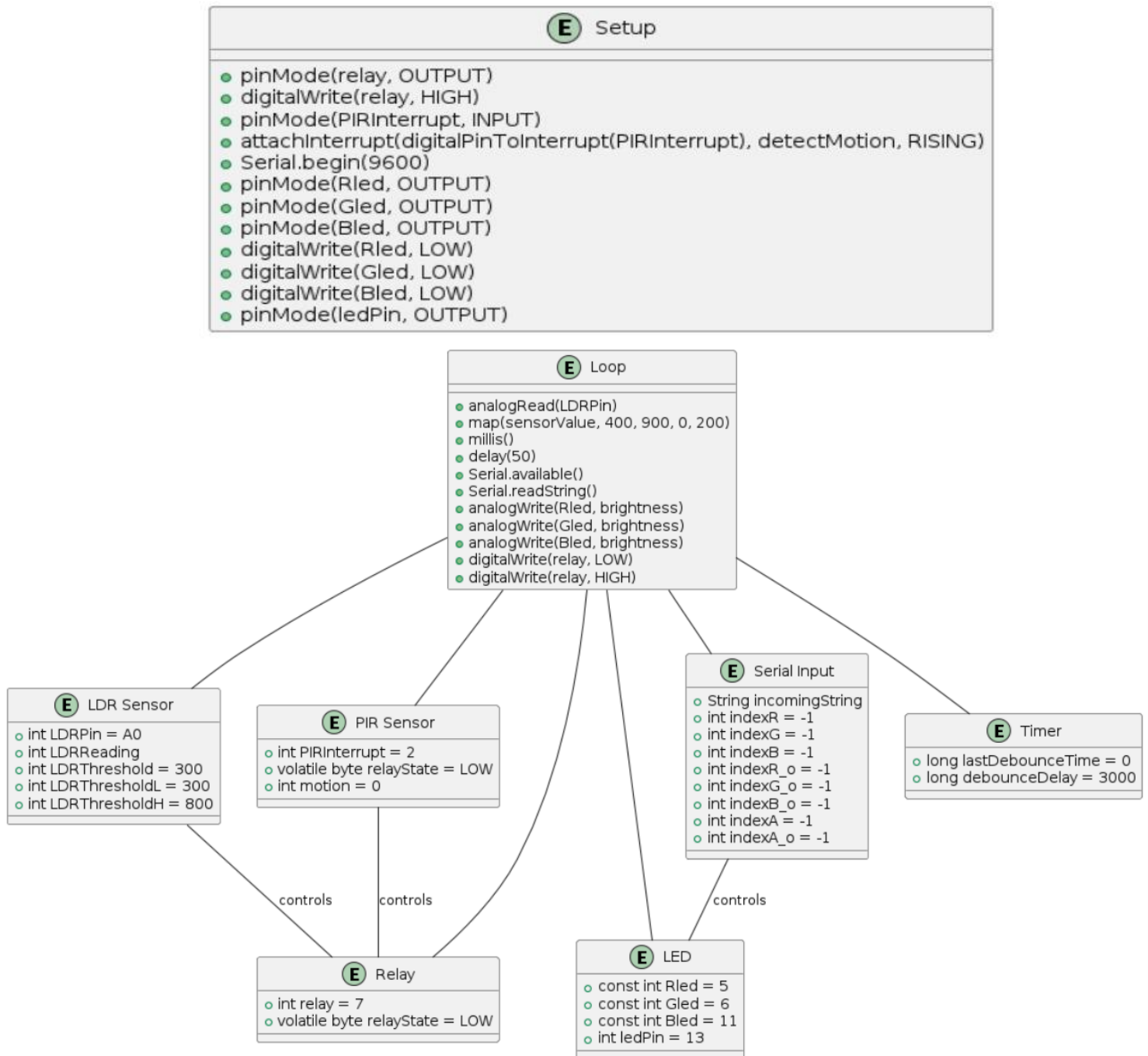


Figure 3.4 ERD Diagram

3.5 Sequence Diagram

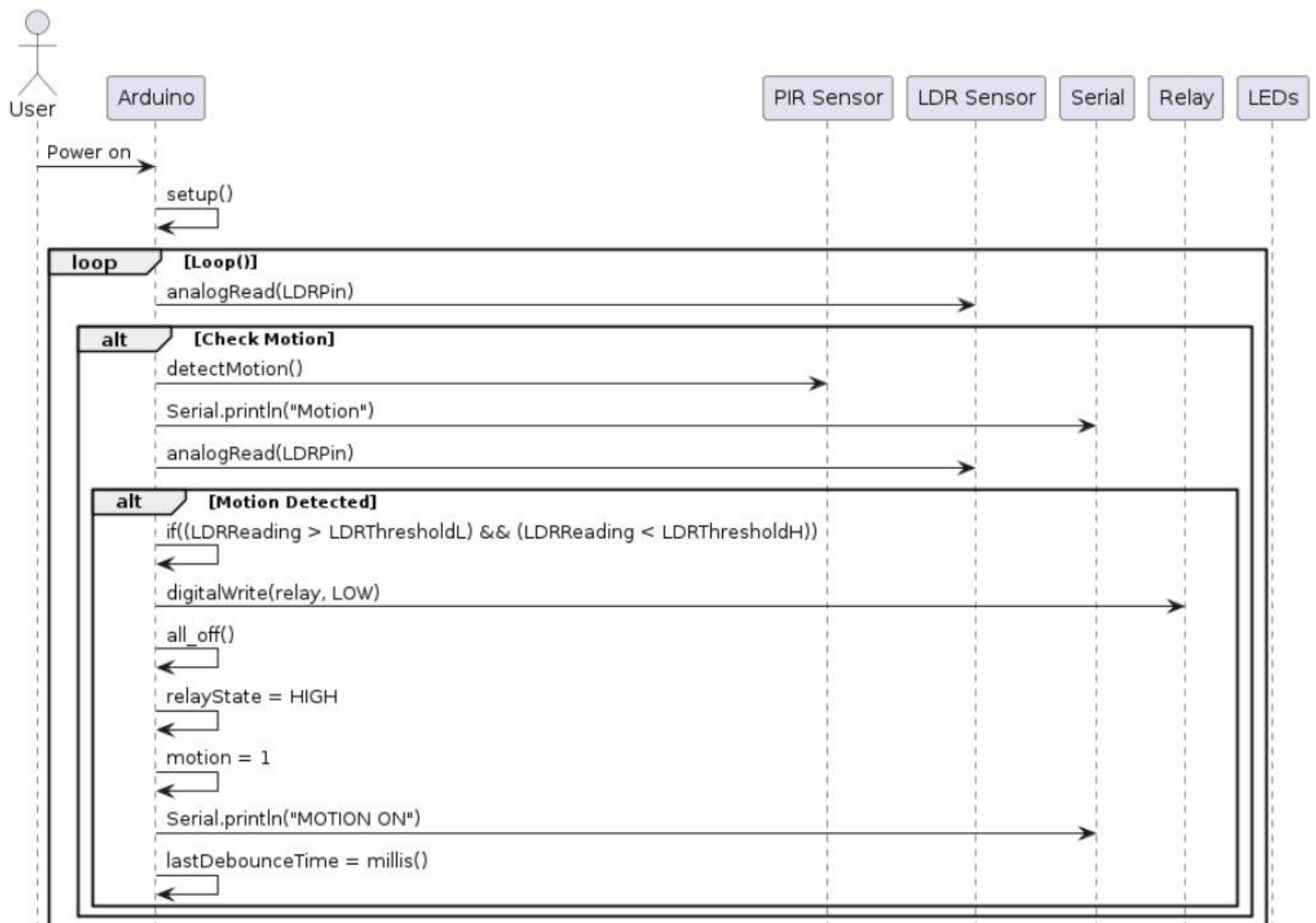


Figure 3.5.1 Sequence Diagram

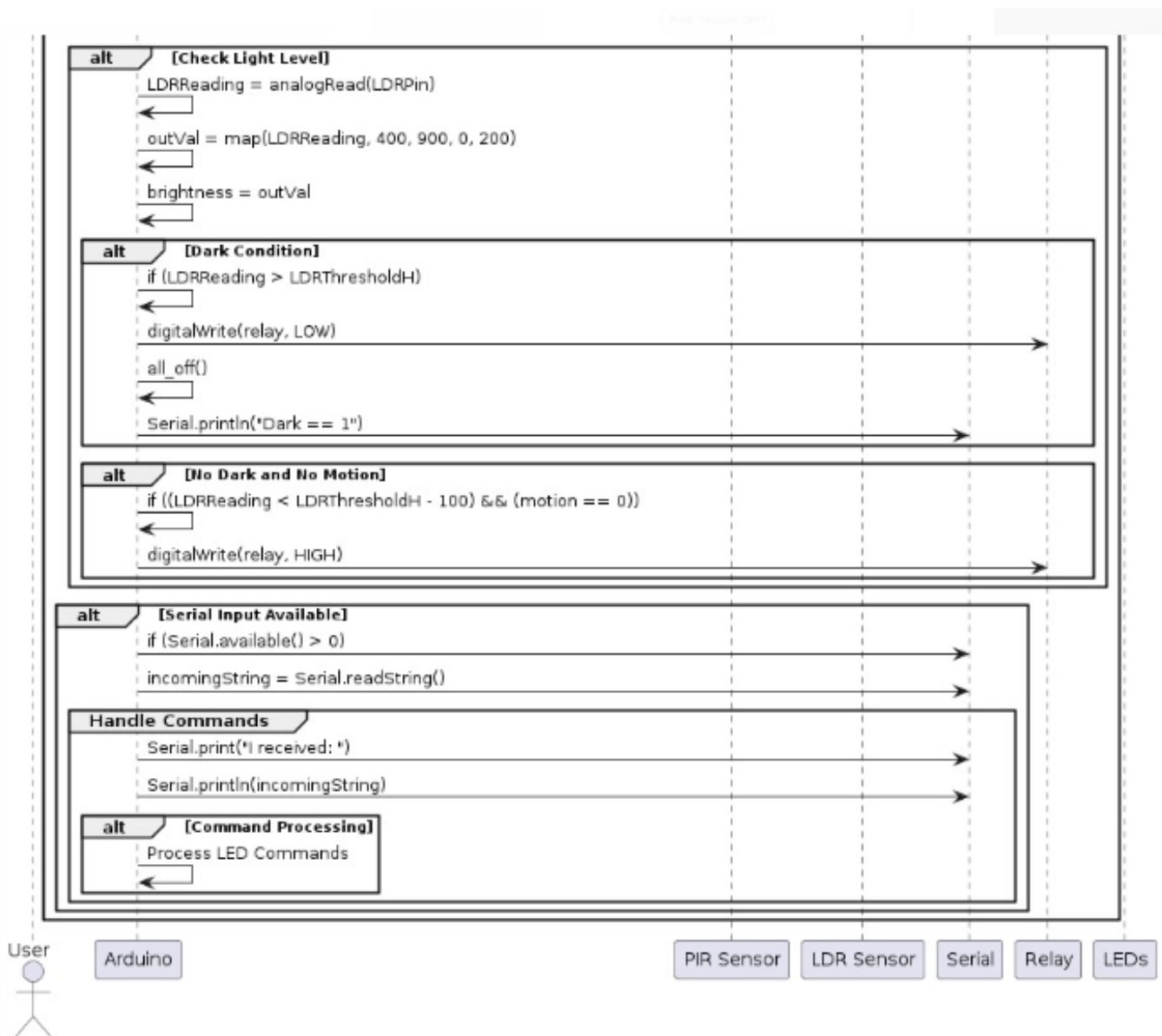


Figure 3.5.2 Sequence Diagram

3.6 Activity Diagram

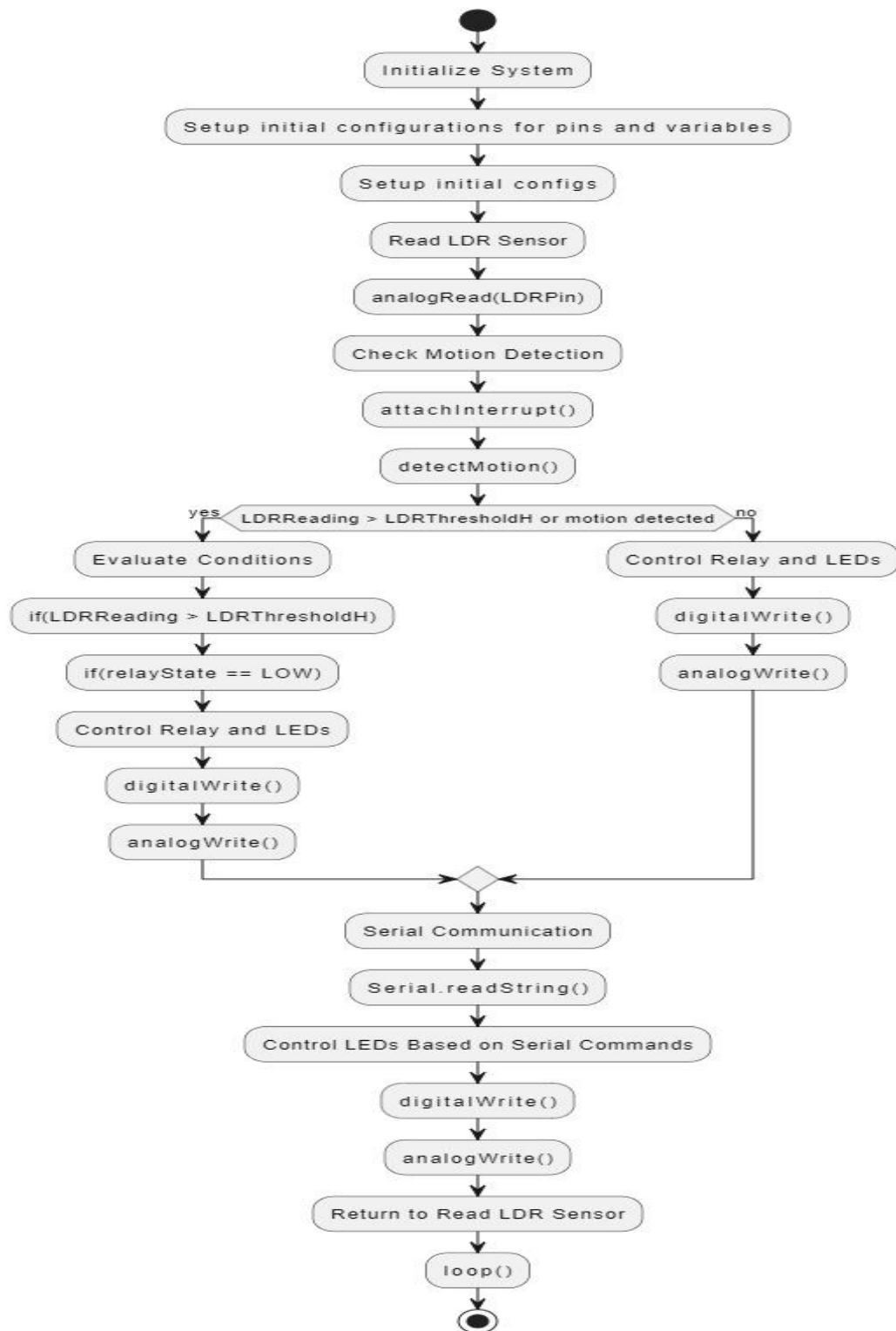


Figure 3.6 Activity Diagram

CHAPTER 4

SYSTEM HARDWARE & SOFTWARE

DESIGN

4.1 Hardware Design

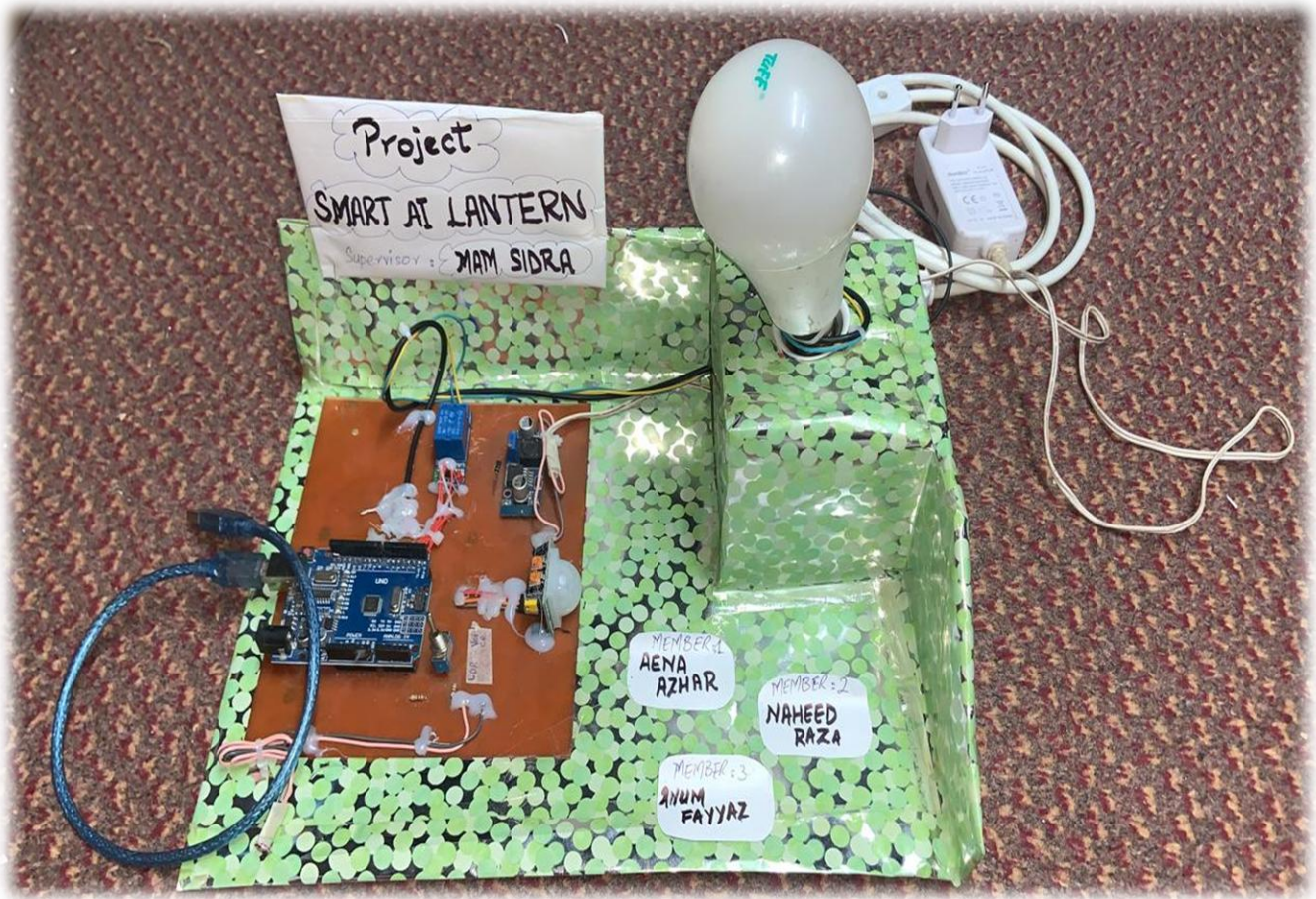


Figure 4.1 Project Picture

1. **Evaluation of hardware additives:** deliver an in depth rundown of each hardware part that is going into the wise AI lantern mild device provide statistics approximately the manufacturer, model numbers, and specs wireless (including sensor types and microcontroller devices).
2. **Block Diagram:** exhibit a block diagram that suggests the overall format of the clever LED lantern gadget. Describe the statistics drift channels, hardware modules (such as sensors and actuators), and their connections.

Example of a Block Diagram:

plaintext --> [PIR Motion Sensor] --> [Microcontroller] [LDR Sensor]

v +--> [Actuators: Relay, RGB LEDs]

4.2 Description Of each hardware components

RGB Bulb

An RGB bulb is a kind of light bulb that can produce various colors through the mixing of red, green and blue lights. Intensities of the LEDs that correspond to these colors must be adjusted in order for them to be created. Multiple ways can be used to control these bulbs including remote controls, smart phone applications or microcontrollers like Arduino.

PIR Motion Sensor

A passive infrared motion sensor (PIR) identifies motion by detecting changes in levels of infrared radiation. It is commonly found in security systems and automatic lighting controls. The detector reacts to heat from living bodies causing it to either turn on an alarm or light when someone moves.

LDR (Light Dependent Resistor)

An LDR also called a photo resistor is a resistor that decreases its resistance with increasing incident light intensity. It finds use in circuits where light sensing is required such as for automatic streetlights and brightness control systems. When it comes into contact with light, its resistance declines and allows more current flow through the circuit.

Relay

A relay is an electromechanical switch employed in controlling high power using low power signals. They enable devices requiring higher voltage and current than those provided by the microcontroller like Arduino's to be controlled by such microcontrollers.

4.3 Circuit Diagrams

4.3.1 RGB Lamp Circuit Diagram

Resistor

|

| |

R| G|

| |

| B|

-|-----|- Ground (Cathode)

| |

| |

| |

D3 D5

| |

.. ..

|| ||

|| ||

' ' ' ' ' '

| |

GND GND

4.3.2 PIR Motion Sensor Circuit Diagram

PIR Sensor

.-----.

| |

|-VCC-OUT-|

|-|---|-|

|-|---|-|

|-GND-----|

|-|-----|

.-----.

|-|-|-|

5V D2 GND

|-|-|-|

Arduino

4.4 Software Design

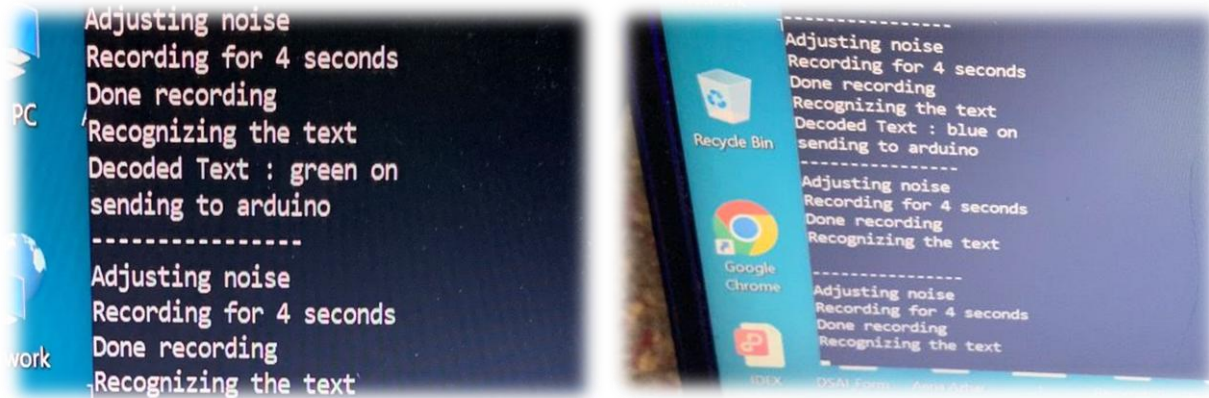


Figure 4.2 Processing Input (a,b)

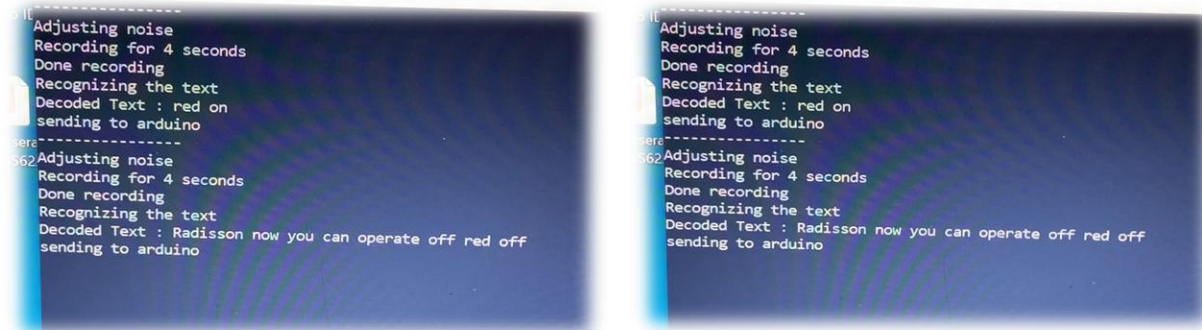


Figure 4.3 Processing Input(c,d)

1. Software program architecture assessment: deliver an outline of the software structure used inside the smart Wi-Fi intelligence lantern lamp device. Describe excessive-degree components such MCU Wi-Fi ware, sensor facts processing algorithms, and actuator manipulate common sense.

2. Description of Firmware, cartoon shape (for Arduino or equivalent):

Describe the setup() and loop() workouts and the general software shape. move over how preset algorithms are used to interpret sensor information, make selections, or

perform actuators.

4.5 Algorithms

Give special wireless about the techniques that have been used (together with the motion detection and light depth manage techniques). Use flowcharts or bits of pseudocode to demonstrate algorithmic reasoning and choice-making processes.

RGB Bulb

Setup:

```
int redPin = 6, greenPin = 3, bluePin=5;
```

Use the setup() function to make these pins as output

Define pins

```
const int redPin = 9;
```

```
const int greenPin = 10;
```

```
const int bluePin = 11;
```

```
void setup() {
```

Set pins as output

```
pinMode(redPin, OUTPUT);
```

```
pinMode(greenPin, OUTPUT);
```

```
pinMode(bluePin, OUTPUT);
```

```
}
```

```

void loop() {

setColor(255, 0, 0); // Red

delay(1000);

setColor(0, 255, 0); // Green

delay(1000);

setColor(0, 0, 255); // Blue

delay(1000);

}

setColor(int red, int green, int blue)

analogWrite(redPin, red);

analogWrite(greenPin, green);

analogWrite(bluePin, blue);

}

```

2. PIR Motion Sensor

Algorithm:

Define the pin is connect to PIR sensor output (e. g., pirPin)

Make this pin Input in setup().

Initialize A Pin to Control a Led (optional)

Loop:

Then, you need to get the sensor value in the loop() block.

Redacted] If motion detected then do something (E. g On LE)

```
pins const int pirPin = 2;
```

```
int ledPin = 13;
```

```
void setup() {
```

```
pinMode(pirPin, INPUT); }
```

```
void loop() if(digitalRead(pirPin) == HIGH)
```

```
{ digitalWrite(ledpin,HIGH);
```

```
}else
```

```
{ digitalWrite(LEDOUTPIN, LOW);
```

```
delay (100);}
```

3. Relay

```
/Define pin
```

```
const int relayPin = 7;
```

```
void setup() {
```

```
pinMode(relayPin, OUTPUT);
```

```
}
```

```
void loop() {  
  
  digitalWrite(relayPin, HIGH);  
  
  delay(5000);  
  
  digitalWrite(relayPin, LOW); // Turn off relay  
  
  delay(5000); // Wait for 5 seconds}
```

- **Microcontroller:** Arduino Uno – ATmega328P, 16MHz, 32KB Flash, 2KB SRAM
- **PIR motion Sensor:** HC-SR501 – Detection range: four–7 meters
- **LDR Sensor:** – Photo resistor – Resistance variety: 10k Ω to 1M Ω

CHAPTER 5
TESTING, RESULTS & DISCUSSION

5.1 Modular Testing

Goals and Range of Testing

Clearly state the testing goals, focusing on the necessity of verifying every system component—hardware modules and embedded software included. Make sure the scope is defined to include scenarios for integrated system testing as well as testing of individual hardware modules.



Figure 5.1 Test Case 1



Figure 5.2 Test Case2

Comprehensive Testing Situations

Hardware Examination:

PIR Motion Detector:

Describe the tests done to confirm the accuracy of motion detection in different scenarios (e.g., distance, angle).

Provide data on reaction times, false positive rates, and detection rates.

LDR Sensor: Describe the experiments carried out to evaluate the accuracy and sensitivity of light levels. Provide information about sensor readings in a range of ambient light intensities.

Actuators (such as relays and RGB LEDs):

Describe the procedures used to assess the efficacy and dependability of actuation (e.g., relay switching, light intensity control).

Give information on the actuator operations' uniformity and responsiveness.

Testing Software:

Functionality of Firmware:

Detailed test cases that are run to verify firmware features such communication protocols, motion detection logic, and light control algorithms. Talk about the tested situations, including how to handle errors (such as malfunctioning sensors or poor communication).

User Interface (if applicable): If there is a user interface, describe the usability studies that were carried out to evaluate configuration settings and user interaction. Share your research's findings about usability and user feedback.

Integration Examination:

Describe the process that was utilized to combine all of the software and hardware pieces into a unified system. Talk about integration test scenarios that evaluate system-wide features such as sensor-based automatic illumination changes.

5.2 Test Results: Results Presentation

Make good use of charts, graphs, and tables when presenting quantitative data. Provide visual displays of actuator responses, sensor data, and system performance indicators (such as energy usage and reaction time).



Figure 5.3 Test Case 3



Figure 5.4 Test Case 4

Assessment and Interpretation

Advantages:

Make some efforts on focusing on the positive working aspects of the system such as door movements detection, energy control efficiency, etc.

Speak about the advantages concerning practical application and the pleasure of the users.

Drawbacks:

Consider any limitations or times when the system failed to work (e. g. , slower reaction rates, challenges with regard to the sensors).

Identify the reasons for poor results and recommend improvements for their enhancement.

5. 3 Results & Discussion

Evaluate the degree to which the system meets the requirements set in the design process (e. g. , performance requirements, users' requirements).

In case applicable, one can discuss the compliance to relevant industry standards or best practices.

Consequences and Prospective Improvements

Think about what these test results mean for its utility, reliability, and readiness for the market of the smart AI lantern lamp.

In light of the strengths and the problems that have been identified, give recommendation for future enhancement or modification.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Project Overview and Summary

1. Project Goals Review: Describe the original issue and the goals declared in the beginning of the introduction. Evaluate the degree to which the suggested solution meets these objectives.

2. Design and Development Process: Briefly describe the smart AI lantern lamp solution, including its design, design development method, key components & features. Focus on the advantages and the uniqueness that your solution has over the existing solutions.

3. Achievement: System Integration Briefly explain the smooth amalgamation of the embedded software, also referred to as firmware and the hardware components including the sensors and actuators.

Discuss how the system effectively leverage on the use of such things as AI algorithms or complex decision-making models.

4. Functionality Validation: Explain how the effectiveness of the motion detection, the level of control over the light intensity, and the user's interaction, among other features of the system, can be confirmed.

Organize focus on successes in the meeting of the users' expectations and the performance parameters.

6.2 Restrictions of the Project

1. Evaluation Method: Explain the approaches applied in the assessment of the smart AI lantern lamp solution. Discuss specific runs, modeling or trials conducted with the intent of validating system performance in concrete situations.

2. Results and Conclusions: True to the nature of a research study that focuses on numbers and facts, this segment provides the outcomes of the testing phase in Chapter 5. When presenting important results such as the happiness of the users, the responsiveness and dependability of the system, one should use a table, chart or graph.

6.3 Conclusion (Limitations, Challenges And Future Recommendations)

1. Restrictions and Difficulties

State and discuss the limitations which have emerged in the testing and development phase (for instance, issues with the sensors' accuracy or challenges with firmware modifications). Think of how these constraints might impact the smart AI lantern lamp's

scalability or use in real life.

2. Future Recommendations: Make concrete suggestions for upcoming upgrades or changes based on lessons discovered and recognized limits. Give top priority to suggestions that deal with important problems that impact system functionality or user experience.

Key Points Synopsis

Describe the key discoveries and accomplishments of the initiative to create a smart AI lantern light. Emphasize how the initial issue statement and the goals listed in Chapter 1 are successfully addressed by the solution.

Importance and Significance

Talk about how the smart AI lantern lamp could affect pertinent fields (such energy efficiency and smart home automation). Emphasize your contributions to improving user quality of life or promoting IoT technology.